



DAFFODIL INTERNATIONAL UNIVERSITY
Department of Computing and Information System (CIS)

Algorithm

Lab Report-04

Submitted By:

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Program: B.Sc. in CIS

Submitted To:

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Class_Code

```
#include<stdio.h>
int main(){
    int a[100],i,j,n,temp;
    printf("Enter Number of element: ");
    scanf("%d",&n);
    printf("Enter element: ");
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(i=0;i<n-1;i++){
        for(j=0;j<n-i-1;j++){
            if(a[j]>a[j+1]){
                temp=a[j];
                a[j]=a[j+1];
                a[j+1]=temp;
            }
        }
    }
    printf("Sorted Array: ");
    for(i=0;i<n;i++){
        printf("%d ",a[i]);
    }
    return 0;
}
```

Output:

```
PS I:\Programming\All Coding> cd "i:\Programming\All Coding\Basic C ,C++\3rd semester
assCode1 } ; if ($?) { .\classCode1 }
Enter Number of element: 4
Enter element: 67
34
23
75
Sorted Array: 23 34 67 75
PS I:\Programming\All Coding\Basic C ,C++\3rd semester Algorithm Lab\Lab report 4> █
```

Task 1

```
#include<stdio.h>
int main(){
    int n, i, j;
    printf("Enter number of students: ");
    scanf("%d", &n);
    int id[n];
    int scr[n];
    int tempScore, tempID;
    printf("Enter Student ID and Score:\n");
    for(i=0;i<n;i++){
        scanf("%d %d",&id[i],&scr[i]);
    }
    for(i=0;i<n-1;i++){
        for(j=0;j<n-i-1;j++){
            if(scr[j]<scr[j+1]){
                tempScore=scr[j];
                scr[j]=scr[j+1];
                scr[j+1]=tempScore;

                tempID=id[j];
                id[j]=id[j+1];
                id[j+1]=tempID;
            }
        }
    }
    printf("\nRank\tStudent ID\tScore\n");
    for(i=0;i<n;i++){
        printf("%d\t%d\t\t%d\n",i + 1,id[i],scr[i]);
    }
    return 0;
}
```

Output:

```
Lab\Lab report 4" ; if ($?) { gcc task1.c -o task1 } ; if ($?) { .\task1 }
Enter number of students: 4
Enter Student ID and Score:
101
80
102
72
103
83
104
90

Rank    Student ID    Score
1       104           90
2       103           83
3       101           80
4       102           72

PS I:\Programming\All Coding\Basic C ,C++\3rd semester Algorithm Lab\Lab report 4>
```

Task 2

```
#include <stdio.h>

#include<string.h>

int main(){

    int n, i, j;
    char name[50][50];
    float price[50];
    printf("Enter number of products: ");
    scanf("%d", &n);
    printf("Enter Product Name and Price:\n");
    for(i = 0; i < n; i++){
        scanf("%s %f", name[i], &price[i]);
    }
    char tempName[50];
    float tempPrice;
    for(i = 0; i < n - 1; i++){
        for(j = 0; j < n - i - 1; j++){
            if(price[j] > price[j + 1]){
                tempPrice = price[j];
                price[j] = price[j + 1];
                price[j + 1] = tempPrice;

                strcpy(tempName, name[j]);
                strcpy(name[j], name[j + 1]);
                strcpy(name[j + 1], tempName);
            }
        }
    }
    printf("\nProducts Sorted by Price\n");
    for(i = 0; i < n; i++){
        printf("%s\t%.0f\n", name[i], price[i]);
    }
    return 0;
}
```

Output:

```
Lab\Lab report 4\ " ; if ($?) { gcc task2.c -o task2 } ; if ($?) { .\task2 }
Enter number of products: 3
Enter Product Name and Price:
mouse
360
SSD
1020
HDD
1000

Products Sorted by Price
mouse    360
HDD      1000
SSD      1020
PS I:\Programming\All Coding\Basic C ,C++\3rd semister Algorithm Lab\Lab report 4> █
```